



CSE110 Sample Final Question-1

Computer Programing (Java-1) (BRAC University)



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No. of Pages	3
No. of Questions	4
Total Marks	30
Time: 80 Minutes	

Department of Computer Science and Engineering
Final Examination
CSE 110: Programming Language I

- ❖ Write theory teacher's Name/Initial on top of the answer script in LARGE FONT.
- ❖ Answer all questions. Use **back part** of the answer script for rough work.
- ❖ Answer Question 1 & 2 at the **beginning part** of the answer script.
- ❖ Figure in bracket [] next to each question indicates marks for that question.
- ❖ At the end of exam, put **question paper** inside answer script and **return both**.
- ❖ Understanding the question is part of the exam, **please do not ask questions**.

No washroom breaks

Section: ____ ID: _____ Name in CAPITAL: _____

Question 1 [CO1] [10 Points]
[Answer on the answer script]

You are given an array containing the names of some planets in our solar system and three additional arrays containing the corresponding **x**, **y**, and **z coordinates** of those planets. Write a Java program that prompts the user to input the name of a planet. The program should then determine and **print** the **closest planet** to the input one along with the **distance** between them. If the user input is not a valid planet name, the program should print "Invalid input". Use the following formula to find distance between two 3D points. [You may use Math.pow() method. You don't need to handle the places after the decimal point in the output.]

$$\text{Distance} = (x_1 - x_0)^2 + (y_1 - y_0)^2 + (z_1 - z_0)^2$$

Given Arrays:

```
String[] planets = {"Mercury", "Venus", "Earth", "Mars", "Jupiter"}
double[] x_coordinates = {0.39, 0.72, 1.00, 1.52, -5.20};
double[] y_coordinates = {0.24, 0.00, 0.00, 0.99, 2.86};
double[] z_coordinates = {-0.10, 0.44, -0.02, 0.21, 0.42};
```

Sample Input 1:

Earth

Sample Output 1:

Closest Planet : Venus

Distance : 0.290000

Explanation:

3D coordinate of earth is = (1.00,0.00,-0.02)
 After calculating the distance with every other planets' coordinates, we get Venus (0.72, 0.00, 0.44) as the closest.
 Distance = $(1-.72)^2 + (0-0)^2 + (-.02-.44)^2 = 0.29$

Sample Input 2:

Pluto

Sample Output 2:

Invalid Input

Explanation:

Since "Pluto" is not found in the given array (planets), the program prints "Invalid Input".

Question 2 [CO1] [10 Points]

[Answer on the answer script]

Write a java program called **StringCutter** that takes three inputs; firstly a **string** and then an **integer** that specifies how many characters to remove from that string. Next a **boolean value** is taken from user that determines where to remove the characters from. The string will be **shortened by the number of characters** provided in the integer input. Now, it will either be cut from the front or the back depending on the boolean input. If the boolean input is **false**, the string will be cut from the **front**. Otherwise the string will be cut from the **back**.

Sample Input 1: Enter string:Programming Enter number of character: 3 From the back? (true/false): false	Sample Output 1: gramming
Sample Input 2: Enter string: Language Enter number of character: 5 From the back? (true/false): true	Sample Output 2: Lan
Sample Input 3: Enter string: Final Exam Enter number of character: 12	Sample Output 3: Number of characters cannot be larger than the String

Question 3 [CO4] [5 Points]

[Answer on the question paper]

The code below is designed to sort an array in **increasing** order. However, it contains **6 errors**. Analyze the code, identify the errors, and correct them. Provide your corrections in the specified format.

1	<code>public class A{</code>
2	<code>public void main(String args []){</code>
3	<code>int array = {20, 25, 10, 8, 3};</code>
4	<code>for (int i = 0; i <= array.length; i++){</code>
5	<code>int min_index = i;</code>
6	<code>for (int j = i+1; j < array.length(); j++){</code>
7	<code>if (array[j] > array[min_index]){</code>
8	<code>min_index = j;</code>
9	<code>}</code>
10	<code>}</code>
11	<code>int temp = array[min_index-1];</code>
12	<code>array[min_index] = array[i];</code>
13	<code>array[i] = temp;</code>
14	<code>}</code>
15	<code>System.out.println("Sorted Array in Increasing Order: ");</code>
16	<code>for (int j = 0; j < array.length; j++){</code>
17	<code>System.out.print(array[j]+" ");</code>
18	<code>}</code>
19	<code>}</code>
20	<code>}</code>

Write your corrections in the table below. For better understanding, one error correction is shown.

Line Number	Fix
Line 2	public static void main (String [] args)

Question 4 [CO1] [6 Points]
[Answer on the question paper]

Illustrate the outputs of the following statements. Provide your workings on the answer script to verify your outputs. **Your answer will not be accepted without the workings. All of the outputs must be in the question paper.**

1	public class methodTracing_A{
2	public static void main(String [] args){
3	mTracing1(8);
4	}
5	public static void mTracing1(int n){
6	if (n>=3){
7	mTracing1(n-1);
8	System.out.println(mTracing2(n));
9	}
10	}
11	public static int mTracing2(int a){
12	int b = (a++) + 4;
13	return a-b*a;
14	}
15	}

Output